

Amirhossien Mahmoudnia

University of Tehran, College of Engineering, School of Surveying and Geospatial Engineering

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Education

 **University of Tehran**
B.Sc. in Surveying and Geospatial Engineering

Sep 2019 – Feb 2024

Selected Courses:

- Fundamentals of Cartography
- Principles of GIS
- Spatial Analysis
- Remote Sensing
- Digital Terrain Modeling

 **University of Tehran**
M.Sc. in GIS - Current GPA: 3.62/4.0

Oct 2024 – Expected Jul 2026

Selected Courses:

- GIS Theories
- Soft computing
- Location-based Services
- Geospatial Web

Research Interest

- GIS | Spatial Analysis
- Geospatial Artificial Intelligence (GeoAI)
- Spatial Optimization
- Environmental Monitoring

Publications

Conference Presentation

Dec 2025

- **Mahmoudnia, A.**, et al., "GeoAI Framework for Evaluation of Deep Learning Models in Water Body Segmentation", ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences.
- **Mahmoudnia, A.**, et al., "Detecting Marsh-to-Open-Water Transitions Using Dual-Date Sentinel-2 and DeepLabV3", ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences.
- **Mahmoudnia, A.**, et al., "GIS-Based Optimization of Humanitarian Aids Logistics in Earthquake-Affected Urban Areas", ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences

Under Review

2026

- **Mahmoudnia, A.**, et al., "Deep Learning Mapping of Wetland Dynamics: A User-Friendly Web-Based Decision Support Tool for Chesapeake Bay", International Journal of River Basin Management (Taylor Francis).
- **Mahmoudnia, A.**, et al., "Frugal GeoAI: Matching Computational Method to Spatial Need", Cartography and Geographic Information Science (Taylor Francis).
- **Mahmoudnia, A.**, et al., "An Uncertainty-Aware Explainable AI Framework for Landslide Susceptibility Mapping Using Dempster-Shafer Evidence Fusion", Stochastic Environmental Research and Risk Assessment (Springer).
- **Mahmoudnia, A.**, et al., "Spatial Optimization of Post-Earthquake Aid Delivery: A Hybrid Genetic Algorithm with Local Search and Clustering Strategy", Transportmetrica A: Transport Science (Taylor Francis)

- Designed and implemented a Spatial Decision Support System (SDSS) for post-earthquake humanitarian aid logistics. Integrated dynamic damage-aware travel time modelling with three hybrid metaheuristic solvers (ACO, Tabu Search, Genetic Algorithm) to generate optimized delivery routes under real-world constraints. The system supports custom user inputs, interactive map visualization, and exportable reports, bridging GIS, combinatorial optimization, and disaster management.

Research Experience

- Developed a Swin-Unet transformer model trained on multi-temporal Sentinel-2 imagery to detect wetland change at Blackwater NWR, achieving $F1 = 0.868$ and mapping 6.32 km^2 of wetland loss over five years, deployed within a Flask-Leaflet decision support dashboard accessible to non-GIS users.
- Benchmarked four state-of-the-art segmentation architectures (U-Net, U-Net++, DeepLabV3+, FPN) under controlled experimental conditions on satellite water body data, establishing DeepLabV3+ as the top performer ($IoU = 0.955$) and delivering actionable model selection guidance for operational remote sensing applications.
- Designed and implemented a Dempster-Shafer evidence fusion framework integrating five XGBoost models across topographic, hydroclimatic, geological, soil, and land-cover data to map landslide susceptibility for Hurricane María, Puerto Rico, achieving ROC-AUC 0.914 and improved probabilistic calibration under spatial cross-validation and external validation; applied SHAP for model interpretability.
- Developed a DeepLabV3-based change detection pipeline on dual-date Sentinel-2 imagery to map marsh degradation at Blackwater NWR, achieving 95.31% overall accuracy and delivering spatially validated conservation priority outputs.
- Developed a reproducible burned area mapping workflow for heterogeneous mountainous terrain in Iran, combining Sentinel-2 spectral differencing with supervised Random Forest classification, achieving near-perfect discrimination ($AUC = 0.998$) between burned and unburned pixels within days of a wildfire event.
- Mapped and predicted LULC change in Tehran across 1990–2030 using multi-temporal Landsat imagery, benchmarking CA-Markov against LCM-MLP and demonstrating that neighbourhood-based transition rules outperform neural approaches for short-term urban expansion projection.
- Solved a large-scale CMDVRP on real crisis-network data from Tehran’s Region 4 using three 2-opt-hybridised metaheuristics (ACO, Tabu Search, Genetic Algorithm) under dual objective functions, demonstrating that hybrid ACO consistently outperforms alternatives by exploiting the asymmetric structure of crisis-time origin-destination matrices.

Coursework Projects

Geospatial Machine learning | Python

Nov 2024 – Jul 2025

- Implemented a K-Nearest Neighbour regression model in Python to estimate spatial distribution of air pollution, leveraging proximity-weighted interpolation to predict pollutant concentrations across unsampled locations from station measurements.
- Applied fuzzy logic to GIS-based spatial analysis to model and propagate uncertainty in geospatial decision-making.
- Applied Recurrent Neural Networks (RNNs) to model hydrological time-series data and forecast flood occurrence using rainfall and river flow records.

Flood Risk Assessment | QGIS, Python, Easyfit, SPSS, Excel

Apr 2025

- Developed a flood hazard assessment model for Mobile County using GIS and remote sensing data, focusing on flood-prone area detection.
- Processed Landsat 8 imagery and Digital Elevation Model (DEM) data in QGIS to create a flood risk index based on elevation and land cover.

- Analyzed a 50-year flood discharge dataset using the Kolmogorov-Smirnov test to assess normality and identified the best-fitting probability distribution using EasyFit.
- Conducted a Regional Flood Frequency Analysis (RFFA) by developing a regression model relating 50-year flood discharge to watershed characteristics, analyzed correlation between variables, and evaluated model performance using statistical metrics.

Remote sensing | *Python, Google Earth Engine, QGIS, ENVI*

Nov 2021 – Aug 2022

- Mapped multi-temporal coastline dynamics using Landsat 8 imagery in Google Earth Engine, extracting shoreline positions across multiple epochs to quantify erosion and accretion patterns along a study area coastline.
- Monitored seasonal water level fluctuations at Layan Dam, Iran, using multi-sensor fusion of Sentinel-1 SAR and Landsat 8 optical imagery, tracking surface water extent changes to characterise intra-annual rise and fall dynamics.
- Computed and compared vegetation and water indices (NDVI, MNDWI, NDWI) across multi-temporal Sentinel-2 imagery to assess seasonal land cover variability and index sensitivity to phenological change.
- Performed land cover change detection using Sentinel-2 optical and SAR data classified by a Random Forest algorithm, identifying temporal changes.

Geovisualization | *HTML/CSS, JavaScript, Cesium, CityGML, ArcGIS Pro, Geo Server*

Oct 2024 – Jul 2025

- Developed a 3D model of an urban area using CityGML, representing buildings, roads, and other infrastructure.
- Developed a web-based 3D spatial visualization platform by converting CityGML to 3D Tiles using FME and integrating it with CesiumJS for interactive geospatial exploration.
- Built a Web-based Augmented Reality (WebAR) application using HTML and JavaScript to overlay geospatial information on real-world landmarks.
- Implemented a Web GIS project to visualize and analyze geospatial datasets interactively using PostGIS and Leaflet.
- Developed a real-time AVL system using Django and Android (Java) to track multiple users' locations, visualize movement on a dynamic map, export routes to GIS formats, and trigger geofence-based alerts.
- Built an Android GPS tracker app with real-time movement visualization on OSM, custom speed calculation, accuracy display, and manual export to GeoJSON and Shapefile formats.

Geoprocessing | *ArcGIS software, Global mapper*

Sep 2021 – May 2022

- Executed core geospatial workflows in ArcGIS including georeferencing, mosaic generation, network analysis, geocoding, and multi-criteria site selection across a series of applied urban and infrastructure mapping scenarios.
- Processed topographic and LiDAR datasets in Global Mapper to generate Digital Terrain Models (DTMs), extract surface features, and conduct segment-level terrain analysis for geospatial applications.
- Integrated multi-layered spatial datasets; population density, land use, and transportation networks, in ArcGIS to produce a cartographic analysis of Esfahan, identifying spatial distribution patterns across urban and rural zones.

Hydrology | *Python, Google Earth Engine, Arc GIS*

Nov 2023 – Dec 2023

- Analysed precipitation patterns and constructed storm hyetographs and hydrographs for a watershed, applying cross-sectional flow analysis to characterize rainfall-runoff response and peak discharge behaviour.
- Integrated satellite-derived hydrological data from Google Earth Engine with ground measurements to improve cumulative precipitation estimates, enhancing the accuracy of rainfall accumulation curves for downstream hydrological modelling.

- Developed a GIS-based model to assess flood risk in a river basin, integrating hydrological data, topography, and historical flood records.
- Delineated watershed boundaries and modelled drainage networks in ArcGIS using DEM-derived flow direction and accumulation layers, supporting water resource assessment and sustainable land management planning.

Photogrammetry

May 2022 – June 2022

- Reviewed and synthesized LiDAR data applications across GIS, urban planning, and BIM frameworks, identifying dominant use patterns in point cloud analysis, feature extraction, and 3D urban modelling, presenting findings to a technical peer audience.
- Executed a full close-range photogrammetry pipeline for a sculptural subject, from acquisition of 100 calibrated images through preprocessing, dense point cloud generation, and mesh reconstruction, producing a photorealistic 3D model visualized in ContextCapture and Metashape.
- Processed 200+ aerial photogrammetry images in Metashape to generate a dense point cloud and high-resolution 3D terrain model of a study area, covering the full pipeline from data preprocessing to final model export.

Work Experience

Internship

July 2022 – Sep 2022

Pey Keian Toos company

Esfarayan, Iran

- Worked as an apprentice land surveyor and gained hands-on experience with production engineering tasks and workflows.
- Created detailed maps using GIS software, incorporating collected GPS data for precise geolocation.

Research Assistant

2023 – Present

 *School of Surveying and Geospatial Engineering, Tehran University — Supervisor: Dr. Zahra Bahramian* *Tehran, Iran*

- Conducted research on geospatial optimization and Spatial Decision Support System development for post-earthquake humanitarian logistics, contributing to two published conference papers, one journal paper under review, and one thesis.

English teacher

Apr 2023 – Aug 2024

C-talk Language Institute

Tehran, Iran

- English mentor for more than 100 students at intermediate and advance level

The 8th ISPRS international Geospatial Conference

Dec 2025

 *School of Surveying and Geospatial Engineering, Tehran University*

Tehran, Iran

- Delivered oral and poster presentation of a peer-reviewed research papers at the 8th ISPRS International Geospatial Conference.

Scientific Society

Sep 2021 – Feb 2024

 *School of Surveying and Geospatial Engineering, Tehran University*

Tehran, Iran

- Member of Scientific Society of Engineering Department at University of Tehran

Technical Skills

Programming/Scripting: Python, MATLAB, HTML/CSS, JavaScript

Tools: Git/GitHub, SQL, Leaflet, PostgreSQL, PostGIS, Geoserver

Softwares: ArcGIS, Revit, Google Earth Engine, Easyfit, SPSS, Global Mapper, QGIS, CityGML

Surveying Instruments: Total Station, GPS

Soft Skills: Teamwork, Project Management, Fast Learner, Problem solving, Initiative, Team Leadership

Languages

Persian: Native

English: Professional working proficiency (TOEFL iBT: Total 98 - Reading 23, Listening 25, Speaking 29, Writing 21)

Hobbies and Interests

- Drawing portrait.
- Coordinating English free discussion sessions at the local language institute.
- Volunteering at Animal Shelters (Regularly volunteer at local animal shelters, helping with the care and adoption process of animals).
- Enjoy story-driven video games with a focus on character development and immersive narratives.

References

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📌 References, Further information, and Proofs are available upon Request.